NBA Career Longevity: What Predicts How Long a Player Lasts in the League?

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Code and data: https://github.com/eylamspivak/statistical-theory-final-project

Abstract— This study investigates what predicts the career longevity of National Basketball Association (NBA) players, using an incident cohort of **2,110** rookies whose debut season is confirmed within the 1997–2022 window of the dataset (441 players whose first *recorded* season coincided with the dataset's own first season were excluded, since a true rookie season could not be confirmed for them). High-scoring rookies had significantly longer careers than low-scoring rookies (Welch $t = 14.06$, $p < 0.0001$, Cohen's $d = 0.61$). Career length differed sharply by draft round (Round-1 mean 7.49 seasons vs. 4.38 for Round-2 vs. 3.22 undrafted; ANOVA $F = 236.8$, $p < 0.0001$, confirmed by Kruskal–Wallis after Levene's test revealed unequal variances). Draft age was negatively associated with career length ($r = -0.294$, $p < 0.0001$). Critically, when rookie-season attributes and draft round were entered into a single nested-model comparison, both contributed statistically independent explanatory power (nested F-tests / GLRT, both $p < 0.0001$): R^2 rose from 0.203 (rookie attributes only) and 0.184 (draft round only) to 0.251 combined — directly confirming, on a modern cohort, the sunk-cost pattern documented by Staw and Hoang [4]. A logistic classifier distinguishing short from long careers, using the same combined feature set, reached **66.1**% accuracy against a **50.7**% majority-class baseline (AUC$= 0.719$). A 1000-simulation Wald sequential test supported the PPG effect in 77.8% of simulations, consistent with the t-test result. Together, these results indicate that both circumstantial factors fixed at draft time and controllable on-court performance independently predict NBA career length.

I. INTRODUCTION

The National Basketball Association (NBA) is the premier professional basketball league in North America, comprising 30 teams that each play an 82-game regular season. Every year, a player's performance is summarized by per-game statistics such as **points per game (PPG)**, **rebounds per game (RPG)**, and **assists per game (APG)** — the average number of points scored, rebounds collected, and passes leading directly to a basket, respectively, across all games a player appeared in that season. A player's first season in the league is referred to as their *rookie season*.

New talent enters the league primarily through the **NBA Draft**, an annual event in which teams take turns selecting eligible players, most commonly from American college basketball or international leagues. Since 1989 the draft has consisted of exactly two rounds of 30 picks each (60 players total) [1]. Being selected in **Round 1** is generally considered a stronger signal of a team's confidence in a player than being selected in **Round 2**, and some eligible players are not selected at all — they remain **undrafted**, and may still join a team afterward, typically with a less secure roster spot. This draft outcome is fixed at the very start of a player's career and is determined by scouts' evaluations of amateur or international performance, not by anything the player has yet done in the NBA itself.

A player's **career length** — the number of distinct NBA seasons in which they appear — is a natural measure of professional success and durability. Career longevity carries practical consequences for players (career earnings, pension eligibility), teams (roster planning, contract risk), and the league more broadly. Yet it is not obvious in advance which factors matter most: does on-court production predict staying power, or do off-court signals fixed at draft time — such as being a high draft pick, or entering the league at a young age — dominate instead?

A. Related Work

This question has a history in sports economics. Staw and Hoang [4] found that draft order predicts NBA playing time and survival *even after controlling for performance*, interpreted as a sunk-cost/escalation-of-commitment effect. Groothuis and Hill [5] confirm this using formal duration (survival) analysis. Fynn and Sonnenschein [6] apply Kaplan–Meier, Cox proportional-hazards, and accelerated-failure-time models directly to NBA career length, finding height and awards extend careers. We revisit this question on a more recent cohort (1997–2022) and, following the sunk-cost literature's core method, explicitly test whether performance and draft position each carry independent explanatory power in the *same* model (Section 2), rather than separately.

In this project we ask: *Can we predict how long an NBA player's career will last based on their rookie-season performance, and which factors — performance-based and circumstantial — drive that outcome?* We break this broad question into four testable hypotheses:

- **H1:** Players who score more points per game in their rookie season have significantly longer careers than lower-scoring players.
- **H2:** Career length differs significantly by draft round (Round 1 vs. Round 2 vs. undrafted).
- **H3:** Age at a player's first NBA season is negatively associated with career length.
- **H4:** Performance (H1) and draft round (H2) each carry explanatory power for career length *independent* of the other,

when tested jointly in the same model (directly following Staw and Hoang [4] and Groothuis and Hill [5]).

B. Dataset

Our data is drawn from the *NBA Players Data* dataset on Kaggle [2], which contains one row per player per season, covering NBA seasons from 1996–97 through 2021–22. Each row includes 21 fields describing that player's performance and biographical information for that season, including points, rebounds, and assists per game; advanced metrics such as net rating, usage percentage, true shooting percentage, and assist percentage; physical attributes (height, weight, age); and draft information (round, pick number, year).

Incident-cohort filtering. The dataset's coverage begins at the 1996–97 season, but many players active in that first season had already begun their careers earlier, outside the data's window. Naively taking each player's earliest *recorded* row as their "rookie season" would misclassify these veterans as young rookies, corrupting age-at-entry and any variable derived from it. We therefore restrict the analytic sample to the *incident cohort*: players whose first recorded season is strictly after 1996–97, for whom we can confirm the recorded first season genuinely was their NBA debut. We report how many players this excludes in Section 2.

From the remaining player-season data we derive two key variables: (i) `career_length`, defined as the span (in seasons) from a player's first to last appearance in the dataset, inclusive — we verified this against the alternative definition of counting only distinct seasons played (which differs only for players with a gap season, e.g. due to injury or an overseas stint) and report the number of affected players in Section 2 — and (ii) each player's *rookie-season* record, obtained by taking their earliest season's row. All subsequent analysis is performed at the player level, using rookie-season statistics to predict eventual career length.

C. Roadmap

Section 2 presents our results in the order the analysis was conducted: from cleaning and describing the data, through testing each hypothesis individually, to a combined regression, nested-model comparison, and classification model with cross-validation. Section 3 details the statistical methods used. Section 4 discusses limitations and directions for further work.

II. RESULTS

A. Data Overview and the Distribution of Career Length

After applying the incident-cohort filter, our sample contains **2,110 players** (**441** players were excluded for having their first recorded season coincide with the dataset's first season, 1996–97), with career length ranging from 1 to 25 seasons (mean = 5.13, median = 4.00, SD = 4.36). As Fig. 1 shows, the distribution is right-skewed (skewness = 1.11). Within this 2,110-player analytic sample, the two `career_length` definitions (span vs. distinct-season-count) differ for **297** players (those with a gap season); we use span throughout for consistency with the paper.

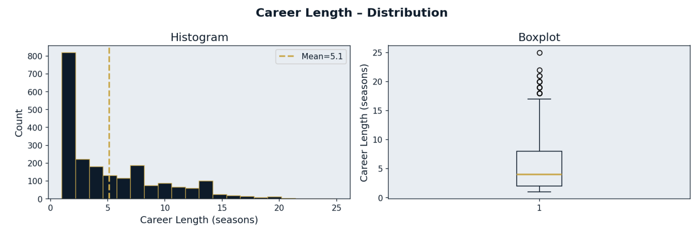

Figure 1: Distribution of career length: histogram (left) and boxplot (right).

2.1.1 Goodness-of-Fit

The Kolmogorov–Smirnov test against a Normal distribution ($D = 0.180$, $p < 0.0001$) and the Shapiro–Wilk test ($W = 0.856$, $p < 0.0001$) both indicate career length is not normally distributed (though the KS result should be treated as indicative only, since its parameters were estimated from the same sample — the Lilliefors problem). Because `career_length` is a small positive integer with many ties, we treat a **discrete χ^2 goodness-of-fit test** (Fig. 3), grouping exact integer values with infinite-tailed end categories, as the primary GOF result: $\chi^2 = 386.5$, $df = 5$, $p < 0.0001$ — normality is decisively rejected. We additionally fit a **Geometric distribution** by MLE ($\hat{p} = 0.195$, implied median survival ≈ 3.2 seasons) as a natural "constant hazard" model for career length, and test its fit with the same grouped χ^2 approach (Fig. 3): $\chi^2 = 73.3$, $df = 6$, $p < 0.0001$ — the geometric model is also rejected. Rather than inferring hazard behavior indirectly from this test alone, Fig. 2 plots the **empirical hazard rate** $\hat{h}(k) = \#\{L = k\}/\#\{L \geq k\}$ directly against the constant hazard the geometric model assumes, and shows exactly why: the hazard follows a **U-shape ("bathtub") pattern**, starting high at $k = 1$ ($\hat{h} = 0.24$, likely players who do not make an NBA roster beyond their rookie deal), declining to a minimum around $k = 5$–7 ($\hat{h} \approx 0.14$–0.15, established players), then rising steadily through the later seasons ($\hat{h} = 0.22$–0.25 by $k = 10$–13, and $\hat{h} = 0.42$ at $k = 14$, though this final point rests on a small risk set of only 132 players and should be treated cautiously). This directly motivates the Kaplan–Meier / hazard-based approach discussed in Section 4.

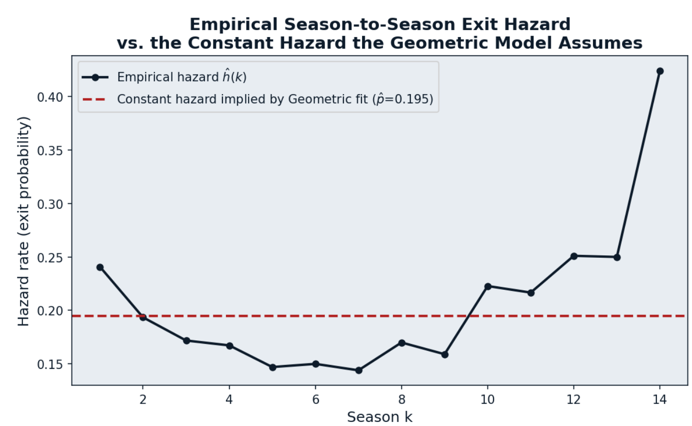

Figure 2: Empirical season-to-season exit hazard vs. the constant hazard implied by the Geometric MLE fit.

B. H1: Rookie Scoring and Career Length

Splitting players at the median rookie-season PPG (3.8), high-scoring rookies ($n = 1064$) averaged **6.39** seasons compared to **3.85** for low-scoring rookies ($n = 1046$). A one-sided **Welch** t-test (unequal variances) confirms this difference: $t = 14.06$,

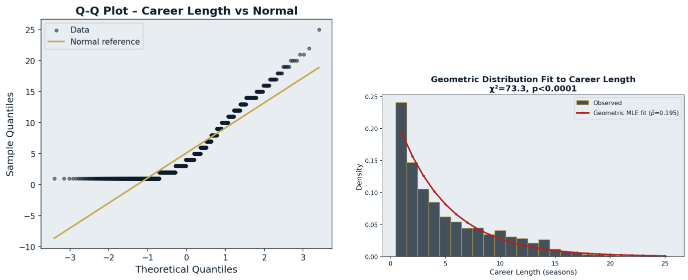

Figure 3: Left: Q-Q plot of career length vs. Normal. Right: Geometric-distribution MLE fit.

$p < 0.0001$, corroborated by the non-parametric Mann–Whitney U test ($p < 0.0001$). The effect size is medium-to-large (Cohen's $d = 0.61$), and the 95% CI for the difference in means, $(2.19, 2.90)$ seasons, excludes zero by a wide margin. Post-hoc power at this observed effect size is > 0.999 (Fig. 5) — this sample is essentially certain to detect an effect this large. **H1 is strongly supported.**

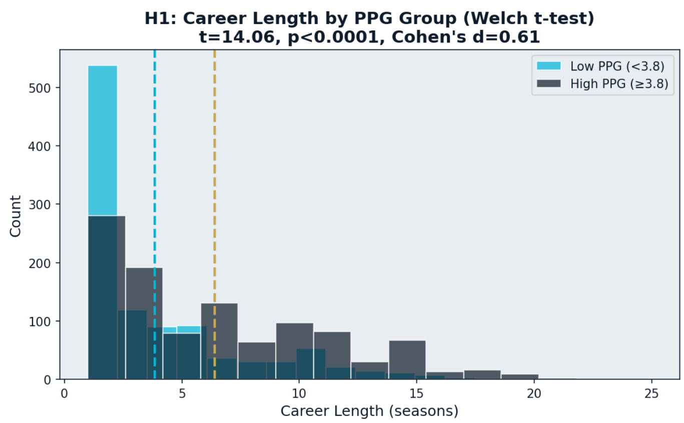

Figure 4: Career length by rookie-season PPG group (Welch t-test).

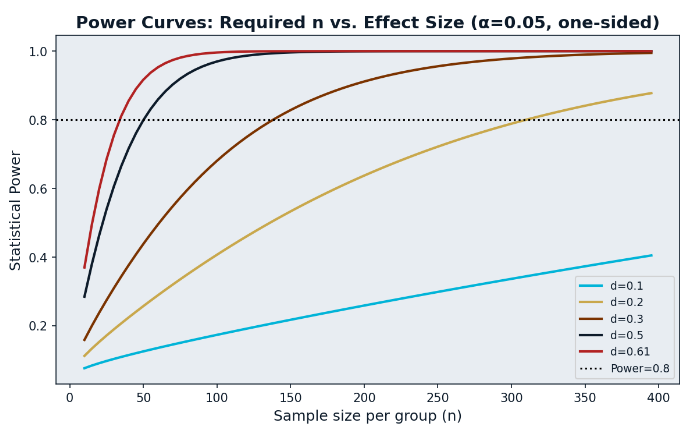

Figure 5: Power curves for the two-sample t-test across effect sizes, showing the sample size needed to reliably detect smaller effects than observed in H1.

C. H2: Draft Round and Career Length

Mean career length was **7.49** seasons for Round-1 picks ($n = 781$), **4.38** for Round-2 picks ($n = 601$), and **3.22** for undrafted players ($n = 728$) (Fig. 6). Levene's test for variance homogeneity ($\text{stat} = 87.07$, $p < 0.0001$) indicates the ANOVA equal-variance assumption is **violated** — group variances differ significantly (Round-1 SD$= 4.63$ vs. Undrafted SD$= 3.18$). A one-way ANOVA nonetheless gives $F = 236.8$, $p < 0.0001$; more importantly, this is corroborated by the non-parametric **Kruskal–Wallis** test ($H = 443.4$, $p < 0.0001$), which does not require the equal-variance assumption and is the more trustworthy result given the Levene violation. Bonferroni-corrected Welch pairwise comparisons ($\alpha = 0.0167$) confirm *all three* pairs differ significantly (Round 1 vs. Round 2: $t = 13.73$, $p < 0.0001$; Round 1 vs. Undrafted: $t = 21.03$, $p < 0.0001$; Round 2 vs. Undrafted: $t = 5.93$, $p < 0.0001$). A Chi-square test of independence between draft group and a three-level career-length category reinforces this: $\chi^2 = 370.3$, $df = 4$, $p < 0.0001$. **H2 is strongly supported.**

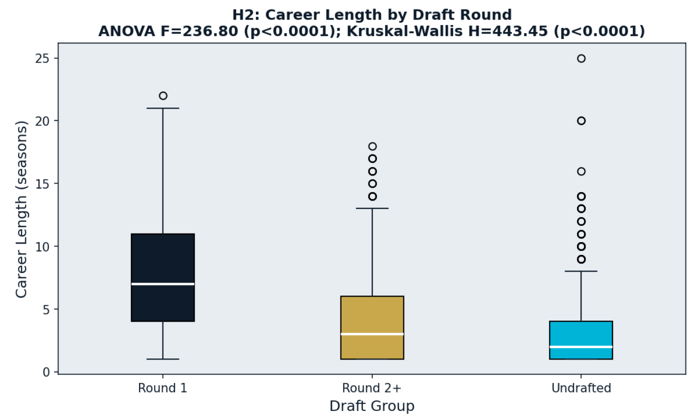

Figure 6: Career length by draft group (ANOVA + Kruskal-Wallis).

D. H3: Draft Age and Career Length

Age at a player's first NBA season shows Pearson $r = -0.294$, $p < 0.0001$ (95% CI $(-0.332, -0.254)$), confirmed by Spearman's $\rho = -0.317$, $p < 0.0001$, in the incident-cohort sample. **Sanity check:** draft-age SD here is **2.40** seasons, close to the true NBA rookie-age SD of ≈1.5–1.7 (see Section 4.1 for the small residual gap and its likely cause).

Sensitivity check: incident vs. prevalent+incident cohort. As a robustness check, we repeated this correlation on the full, unfiltered sample (prevalent + incident cohorts, $n = 2{,}551$, i.e. including players already active before the dataset's 1996–97 start). The association is weaker there ($r = -0.210$) than in the incident cohort alone ($r = -0.294$). This is consistent with **length-biased sampling** in the prevalent-cohort component: a cross-sectional data window over-represents already-established, long-career players among those whose careers began before the window opened, which dilutes the observed age–longevity association. We use the incident cohort as our primary analytic sample throughout this paper for this reason. Fig. 7 shows the incident-cohort relationship (OLS slope $= -0.53$). **H3 is supported.**

E. Multiple Regression: Combining All Predictors

We fit an OLS regression of career length on eight standardized rookie-season predictors (Fig. 8): $R^2 = 0.203$ (Adj-$R^2 = 0.200$). Five predictors are significant at $p < 0.05$: RPG ($\beta = 0.83$), APG ($\beta = 0.53$), draft age ($\beta = -0.94$), usage % ($\beta = 0.24$, $p = 0.011$), and PPG ($\beta = 0.35$, $p = 0.047$, only marginally significant despite its strong univariate H1 effect — likely due to collinearity with RPG/APG, $r = 0.65$–0.73; see Fig. 14). Net rating narrowly misses significance ($p = 0.056$); height/weight are not significant. Draft age remains the largest-magnitude coefficient, comparable to RPG (see the incident-

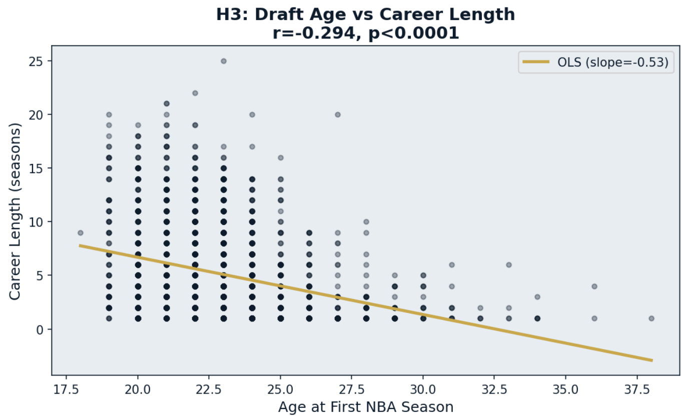

Figure 7: Career length vs. age at first NBA season.

cohort sensitivity check in Section 2.D for why this coefficient is not as extreme as it would be on an unfiltered sample).

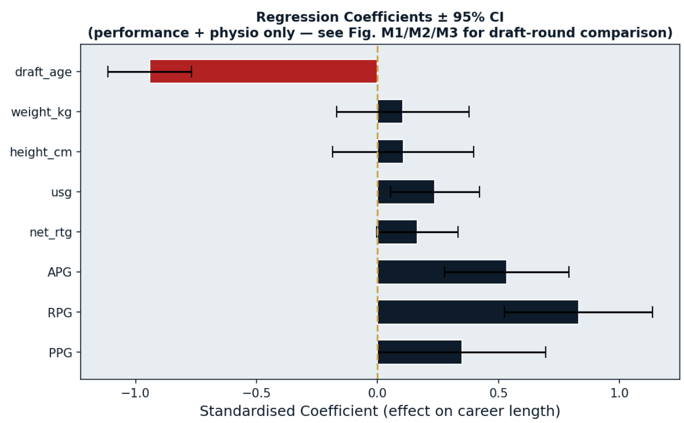

Figure 8: Standardized OLS coefficients (performance + physio only) with 95% CIs.

F. H4: Nested Model Comparison — Rookie Attributes vs. Draft Round

Following Staw and Hoang [4] and Groothuis and Hill [5], we test whether on-court performance and draft round each carry independent explanatory power by comparing three nested models (Fig. 9): **M1** (rookie-season attributes — performance, physical measurements, and draft age together, $R^2 = 0.203$), **M2** (draft group only, $R^2 = 0.184$), and **M3** (both, $R^2 = 0.251$). The nested F-test from M1 to M3 — algebraically a Generalized Likelihood Ratio Test for nested linear models (Section 3) — tests whether draft group adds power beyond rookie-season attributes: $F(2, 2099) = 66.7$, $p < 0.0001$ (equivalently, GLRT $-2 \log \Lambda = 130.0 \sim \chi^2_2$, $p < 0.0001$). The reverse test (M2 to M3) tests whether rookie-season attributes add power beyond draft group: $F(8, 2099) = 23.6$, $p < 0.0001$. **Both directions are highly significant**: rookie-season attributes and draft round each carry statistically independent explanatory power, neither subsuming the other. This directly confirms **H4** and replicates, on a modern (1997–2022) cohort, the core finding of Staw and Hoang [4] and Groothuis and Hill [5] that draft position affects career outcomes even after controlling for on-court performance. Note that M1 already includes draft age alongside performance and physical measurements; the interaction analysis below and the regression in the previous subsection isolate the specific contribution of scoring output (PPG) within this broader set.

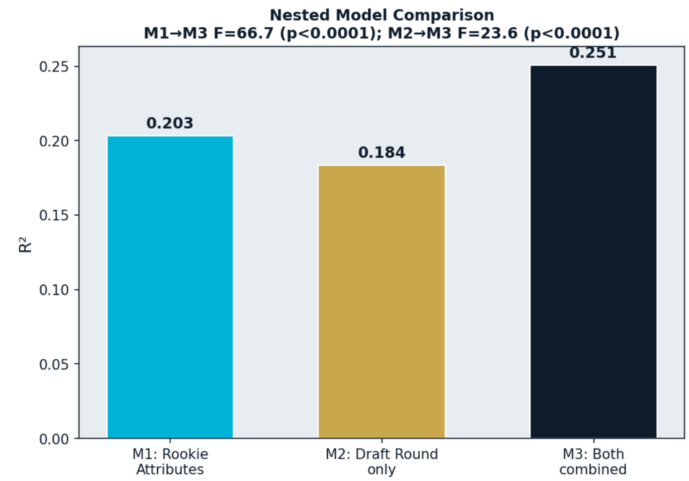

Figure 9: Nested model comparison: R^2 for rookie-attributes-only, draft-only, and combined models.

G. Does Performance Compensate for Draft Position? An Interaction Effect

Adding an interaction term between rookie PPG and draft group (Fig. 10) tests whether PPG's effect on career length depends on draft round. The PPG×Round2+ interaction is significant ($\beta = 0.71$, $p = 0.0057$): the slope relating PPG to career length is steeper for Round-2+ picks than for Round-1 picks. The PPG×Undrafted interaction is **not** significant ($\beta = -0.07$, $p = 0.77$). We do not interpret this as evidence that performance fails to help undrafted players reach longer careers — only **30 undrafted players** in the sample have a standardized PPG above 1 (i.e., reach high scoring output at all), so this cell is too thin to estimate the interaction precisely. The correct reading is that our data cannot distinguish a true null effect from a low-power result for this subgroup, not that performance is proven irrelevant for undrafted players.

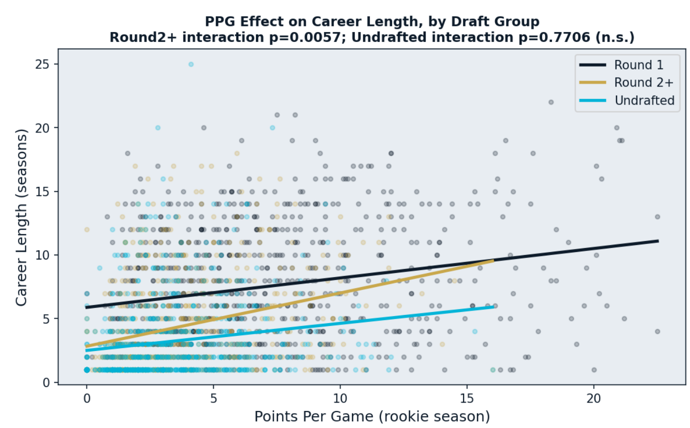

Figure 10: Fitted PPG–career length slopes by draft group.

H. Classification: Predicting Long vs. Short Careers

A logistic regression on the same feature set as M3 (rookie-season attributes plus draft-group dummies), evaluated on a held-out 25% test set, achieved **66.1% accuracy** and **AUC**$= 0.719$ (Fig. 11, confusion matrix Fig. 11) against a **50.7%** majority-class baseline (classes were nearly balanced: 50.7% Long vs. 49.3% Short) — a meaningful improvement over chance, consistent with H4: draft round carries additional, independent predictive signal beyond performance and physical attributes alone. All coefficients point in the expected direction: PPG, RPG, APG, net rating, usage, height, and weight all increase the odds of a long career; draft age, being drafted in Round 2 or later ($\beta = -0.45$), and going undrafted ($\beta = -0.67$, the largest- magnitude coefficient in the model) all decrease it.

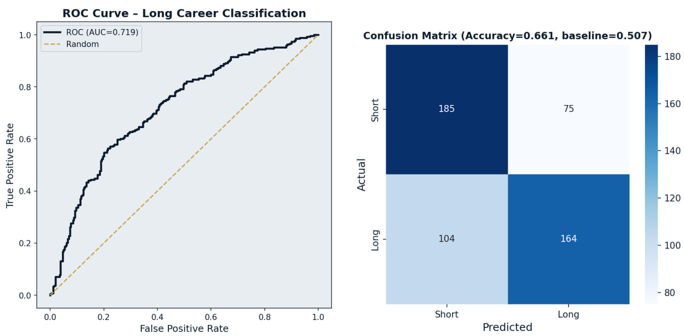

Figure 11: Long-vs-Short career classifier (M3 feature set): ROC curve (left) and confusion matrix (right).

I. Model Validation: Cross-Validation

5-fold cross-validation of **M3** (rookie-season attributes plus draft-group dummies — the paper's headline model, rather than M1 alone) (Fig. 12) gives mean CV $R^2 = 0.238$ (± 0.032), compared to in-sample $R^2 = 0.251$ (matching M3's value in Section 2.G exactly, as expected) — a gap of only 0.013. This small gap indicates the model **generalizes well** and is not meaningfully overfit.

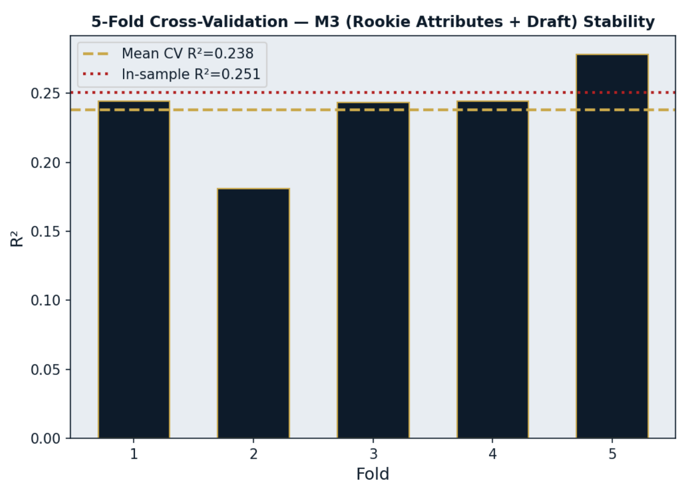

Figure 12: 5-fold cross-validated R^2 vs. in-sample R^2 for M3.

J. Sequential Testing: Wald SPRT (1000-Simulation)

Because a single sequence's stopping decision depends on the order in which observations arrive, we run the Wald SPRT across 1000 independent random re-orderings of the high-PPG subgroup rather than a single pass, testing $H_0 : \mu = 5.13$ (overall mean) against $H_1 : \mu = 6.44$ (0.3σ above baseline; the subgroup's true mean, 6.39, lies close to H_1) (Fig. 13). Across 1000 simulations, the test **rejects H_0 (supports H_1) in 77.8%** of runs and accepts H_0 in 22.2%, with no "no-decision" cases, consistent with H1's much larger-sample t-test finding above. The empirical mean stopping time (ASN) is **48.1** observations (median 38), reasonably close to Wald's theoretical ASN approximation of **42.4**.

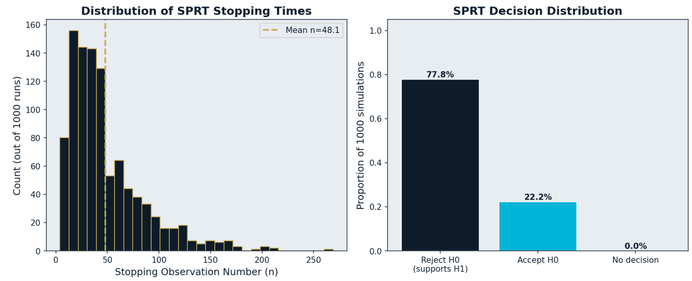

Figure 13: Distribution of SPRT stopping times and decisions across 1000 random re-orderings.

K. Correlation Structure

Fig. 14 summarizes pairwise correlations among key variables. PPG, RPG, and APG are strongly inter-correlated ($r = 0.65$–0.73), as expected since high-usage players tend to accumulate multiple counting statistics simultaneously — this collinearity is the most likely explanation for PPG's comparatively weak, only-marginal significance in the multiple regression (Fig. 8) despite its strong univariate effect in H1: once RPG and APG are in the model, PPG has limited additional (non-shared) variance left to explain. Draft age is negatively correlated with all three performance statistics, height, and career length itself, consistent with the H3 and H4 results above.

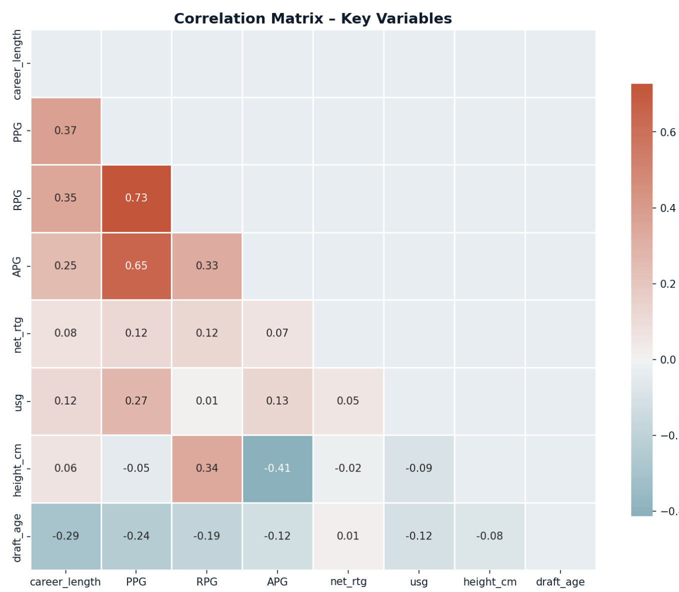

Figure 14: Correlation matrix of key variables.

III. METHODS

A. Q-Q Plot

As a visual complement to the formal goodness-of-fit tests above, we use a Quantile-Quantile (Q-Q) plot, which plots the sample's ordered values against the quantiles of a theoretical reference distribution [3]. Points falling along the $y = x$ reference line indicate agreement with the reference distribution; systematic curvature indicates the specific way the sample's distribution departs from it (e.g. heavier tails, skew).

B. Goodness-of-Fit: KS, Shapiro–Wilk, and Discrete χ^2

The Kolmogorov–Smirnov (KS) test evaluates whether a sample is drawn from a specified theoretical distribution by comparing the empirical cumulative distribution function (ECDF) $F_n(x)$ of the sample to the cumulative distribution function $F(x)$ of the reference distribution: $D_n = \sup_x |F_n(x) - F(x)|$. A subtlety applies when the reference distribution's parameters (e.g. μ, σ) are estimated from the *same* sample being tested: the standard KS null distribution is then no longer exact (the Lilliefors problem), so we report KS results as indicative only. The **Shapiro–Wilk** test is a more powerful alternative for detecting non-normality. Because `career_length` is a small positive integer with many tied values, violating KS's continuity assumption, we treat a **discrete Pearson χ^2 goodness-of-fit test** as primary: observations are binned into k categories, and

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1-m},$$

where E_i is the expected count in bin i under the reference distribution and m is the number of parameters estimated from the data. We apply this both against a Normal reference and against a **Geometric distribution** fit by maximum likelihood ($\hat{p} = 1/\bar{X}$, the natural "constant season-to-season exit probability" model for career length, $P(L = k) = (1-p)^{k-1}p$).

C. Independent-Samples Welch t-test and Mann–Whitney U Test

To compare mean career length between two independent groups (e.g., high- vs. low-scoring rookies), we use **Welch's t-test**, which does not assume equal variances between groups:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

where $\bar{X}_i$, s_i^2, and n_i are the sample mean, variance, and size of group i; this also matches the standard error used for our confidence intervals below. We use Welch's version rather than the equal-variance (Student's) t-test because group variances differ between our draft-round subgroups (confirmed by Levene's test, Section 3.D). The t-test still assumes approximately normal data; because our goodness-of-fit tests indicate career length is right-skewed, we pair every t-test with the non-parametric **Mann–Whitney U test** as a robustness check. The Mann–Whitney test ranks all observations from both groups together and computes

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1,$$

where R_1 is the sum of ranks in group 1 (and U_2 analogously for group 2), testing whether one group tends to produce larger values than the other without assuming any particular distribution.

D. Effect Size and Statistical Power

Statistical significance (p-value) does not by itself convey the practical magnitude of a difference, nor the test's ability to detect one. We report **Cohen's d**,

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}}, \qquad s_{\text{pooled}} = \sqrt{\frac{s_1^2 + s_2^2}{2}},$$

which expresses the gap between two group means in units of pooled standard deviation ($|d| \approx 0.2, 0.5, 0.8$: small, medium, large). We additionally report **statistical power**: $1 - \beta$, the probability of correctly rejecting H_0 when a real effect of a given size exists, computed exactly via the noncentral t-distribution with noncentrality parameter $\delta = d\sqrt{n/2}$ (equal group sizes). This connects directly to the trade-off between Type I error (α, falsely rejecting a true H_0) and Type II error (β, failing to reject a false H_0): for fixed α, power increases with sample size and with the true effect size, which we illustrate with a power curve across a range of n and d (Fig. 5).

E. Variance Homogeneity: Levene's Test, and Kruskal–Wallis

One-way ANOVA (below) assumes equal variances across groups. **Levene's test** checks this by testing whether the group means of $|X_{ij} - \bar{X}_i|$ (absolute deviations from each group mean) are themselves equal across groups, via a standard one-way ANOVA on those deviations; a significant result indicates heteroscedasticity. As a non-parametric counterpart to ANOVA — filling the same role Mann–Whitney fills for the two-sample t-test — we also report the **Kruskal–Wallis H test**, which extends the Mann–Whitney rank-sum idea to more than two groups without assuming equal variances or normality:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1),$$

where R_i is the sum of ranks in group i (ranking all N observations together) and n_i is group i's size; under H_0, $H \sim \chi^2_{k-1}$ approximately.

F. One-Way ANOVA and Bonferroni Post-Hoc Correction

To compare mean career length across more than two groups (Round 1, Round 2+, and Undrafted players), we use one-way Analysis of Variance (ANOVA), which tests $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ via the F-statistic, the ratio of between-group to within-group variance. A significant ANOVA result indicates *at least one* group differs, but not which, so we follow it with pairwise Welch t-tests, applying the **Bonferroni correction** (testing each pair at α/m, m=number of comparisons) to control the family-wise error rate (FWER). Our study runs ≈25 tests in total; we apply Bonferroni locally where multiplicity is explicit (draft-round pairs) and treat any borderline ($0.01 < p < 0.05$) result elsewhere with appropriate caution given this broader FWER concern.

G. Correlation: Pearson and Spearman

To quantify the linear association between two continuous variables (e.g., draft age and career length), we compute the **Pearson correlation coefficient**,

$$r_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y},$$

together with a 95% confidence interval obtained via the Fisher z-transformation, $z = \tanh^{-1}(r)$, which is approximately normally distributed with standard error $1/\sqrt{n-3}$. Because Pearson's r assumes a linear relationship and is sensitive to outliers, we additionally report **Spearman's rank correlation ρ**, computed identically to Pearson's r but on the ranks of the data rather than the raw values, making it robust to monotonic non-linear relationships.

H. Multiple Linear Regression (OLS)

To model career length as a function of several rookie-season variables simultaneously, we fit an Ordinary Least Squares (OLS) regression,

$$y = X\beta + \varepsilon,$$

where y is the vector of career lengths, X is the design matrix of standardized predictors (points, rebounds, and assists per game, net rating, usage percentage, height, weight, and draft age), and $\hat{\beta} = (X^\top X)^{-1} X^\top y$ minimizes the sum of squared residuals. Model fit is summarized by $R^2 = 1 - \mathrm{SS}_{res}/\mathrm{SS}_{tot}$, the proportion of variance in career length explained by the model, and each coefficient's significance is assessed via its t-statistic, $t_j = \hat{\beta}_j / \mathrm{SE}(\hat{\beta}_j)$.

I. Nested Model Comparison and the Generalized Likelihood Ratio Test (GLRT)

To directly test H4 — whether rookie-season attributes and draft round each carry *independent* explanatory power, rather than one subsuming the other — we compare three nested OLS models: rookie-season attributes only (M1: performance, physical measurements, and draft age), draft-group-only (M2), and both combined (M3). For nested linear models under normally-distributed errors, the classical nested F-test,

$$F = \frac{(\mathrm{SS}_{res,\,reduced} - \mathrm{SS}_{res,\,full})/q}{\mathrm{SS}_{res,\,full}/(n - k_{full} - 1)} \sim F_{q,\,n-k_{full}-1},$$

(where q is the number of additional parameters in the full model) is algebraically equivalent to a **Generalized Likelihood Ratio Test (GLRT)** comparing the reduced and full models: the GLRT statistic $-2\log\Lambda = n\log(\mathrm{SS}_{res,\,reduced}/\mathrm{SS}_{res,\,full})$ is asymptotically χ^2_q-distributed under H_0 (reduced model is adequate), and gives the same substantive conclusion as the F-test. We report both forms. Testing M1→M3 asks whether draft group improves the model beyond rookie-season attributes alone; testing M2→M3 asks the reverse.

J. Neyman–Pearson Optimality: MP and UMP Tests

For simple hypotheses $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$, the Neyman–Pearson lemma shows the likelihood-ratio test is **Most Powerful (MP)** at any significance level — precisely the rule underlying the Wald SPRT below, applied sequentially. For the composite alternative $H_1 : \mu > \mu_0$ used in our t-tests, the one-sided t-test is **Uniformly Most Powerful (UMP)** among unbiased tests. This is why t-tests (and their Welch variant) are our primary tool for directional hypotheses, with Mann–Whitney as a distribution-free check.

K. Regression with Interaction Terms

A purely additive regression assumes each predictor's effect on career length is constant regardless of the other predictors' values. To test whether the effect of rookie PPG on career length *depends* on draft round, we extend the model with an interaction term,

$$y = \beta_0 + \beta_1\mathrm{PPG} + \beta_2 D + \beta_3(\mathrm{PPG} \times D) + \varepsilon,$$

where D is a dummy variable for draft group. A significant β_3 indicates that the slope relating PPG to career length differs across draft groups — for example, that strong rookie performance compensates more for a low draft position than for a high one, or vice versa.

L. Chi-Square Test of Independence

To test whether two categorical variables (draft group and a career-length category of Short/Medium/Long) are statistically independent, we construct a contingency table of observed counts O_{ij} and compute

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where E_{ij} is the expected count under independence. A large χ^2 relative to its degrees of freedom (and correspondingly small p-value) indicates the two variables are associated rather than independent.

M. Wald Sequential Probability Ratio Test (SPRT)

Unlike the fixed-sample tests above, the Wald Sequential Test evaluates evidence *as observations arrive*, stopping as soon as enough evidence has accumulated to decide between two simple hypotheses $H_0 : \mu = \mu_0$ and $H_1 : \mu = \mu_1$ (by the Neyman–Pearson/MP argument above, the per-step log-likelihood-ratio is the optimal statistic to accumulate). For two Normal distributions with common variance σ^2, the exact per-observation increment is

$$\log\frac{f(x \mid H_1)}{f(x \mid H_0)} = \frac{\mu_1 - \mu_0}{\sigma^2}\,x \;-\; \frac{\mu_1^2 - \mu_0^2}{2\sigma^2},$$

and the cumulative sum $\Lambda_n = \sum_{i=1}^{n} \log[f(x_i \mid H_1)/f(x_i \mid H_0)]$ is compared against two boundaries, $A = \log\bigl(\frac{1-\beta}{\alpha}\bigr)$ and $B = \log\bigl(\frac{\beta}{1-\alpha}\bigr)$, determined by the desired Type I error rate α and Type II error rate β. Sampling stops and H_0 is rejected once $\Lambda_n \geq A$, or accepted once $\Lambda_n \leq B$; otherwise, sampling continues. Because a single sequence's stopping point depends on the order in which observations arrive, we run the procedure across 1000 independent random re-orderings of the data and report the distribution of decisions and stopping times, together with Wald's classical approximation for the Average Sample Number (ASN), $\mathrm{E}[N \mid H_1] \approx \bigl[(1-\beta)A + \beta B\bigr] / \mathrm{E}_{H_1}[\log(f(x \mid H_1)/f(x \mid H_0))]$, as a theoretical cross-check on the empirical mean stopping time.

N. Logistic Regression for Classification

To predict a binary outcome — whether a player's career is *Long* (at or above the median career length) or *Short* — we use logistic regression, which models the log-odds of the positive class as a linear function of the predictors,

$$\log\frac{P(y = 1 \mid X)}{1 - P(y = 1 \mid X)} = X\beta.$$

Coefficients are estimated by maximum likelihood. Model performance is evaluated on a held-out test set (25% of players, stratified by class) using **accuracy**, the proportion of correctly classified players, compared against the **majority-class baseline** (the accuracy achievable by always predicting the more common class — the correct comparison point, rather than a naive 50%, whenever classes are not perfectly balanced); the **confusion matrix**, cross-tabulating predicted vs. actual class; and the **ROC**

curve and its area under the curve (**AUC**), which summarize the trade-off between true- and false-positive rates across all classification thresholds. An AUC of 0.5 indicates no better than random guessing, while 1.0 indicates perfect separation.

O. k-Fold Cross-Validation

Reporting R^2 from a model fit and evaluated on the same data risks overstating performance due to overfitting. We therefore assess generalizability with k-fold cross-validation ($k = 5$): the data is randomly partitioned into five equal folds; the model is trained on four folds and evaluated on the held-out fifth, repeated so that each fold serves as the test set exactly once. The five resulting R^2 values are averaged, and a large gap between in-sample R^2 and mean cross-validated R^2 signals overfitting.

IV. DISCUSSION

All four hypotheses were supported. Rookie-season scoring (H1) and draft round (H2) both showed large, highly significant effects; age at entry (H3) showed a smaller but robust effect; and, critically, H4 confirms performance and draft round each carry **independent** explanatory power (nested F-tests, both $p < 0.0001$) — neither subsumes the other. Rather than asking which single factor "wins" in isolation, H4 shows both matter *jointly*, replicating Staw and Hoang [4] and Groothuis and Hill [5] on a modern cohort.

The interaction result adds a nuance worth stating carefully: strong rookie scoring predicts a steeper career-length benefit for Round-2+ picks than Round-1 picks ($p = 0.0057$), but we found no significant interaction for undrafted players ($p = 0.77$) — not because performance is irrelevant for them, but because only 30 undrafted players in our sample reach high rookie PPG, leaving too little data to estimate that interaction precisely. This distinction (a genuine null vs. a low-power result) should not be glossed over.

Model validation supports these conclusions: the small gap between M3's in-sample and cross-validated R^2 (0.251 vs. 0.238) indicates the regression is not meaningfully overfit, and the M3-feature-set classifier's 66.1% accuracy against a 50.7% majority-class baseline (AUC$=$ 0.719) confirms that rookie-season attributes and draft-group information together carry genuine predictive signal.

A. Limitations

The most significant remaining limitation is **right-censoring**: players whose careers were still active near the end of the dataset's coverage (the 2021–22 season) have their career length recorded as if it were already final, when in reality their careers may have continued beyond the data's collection window. This affects recently-drafted players disproportionately and likely biases their career-length estimates downward. Standard OLS and logistic regression, as used here, do not account for this censoring; this is distinct from, though related to, the left-truncation issue corrected above (which concerned careers starting *before* the data window, whereas censoring concerns careers continuing *after* it).

A second limitation is that the regression explains only a modest share of the variance in career length, meaning most of what determines how long a player's career lasts — injuries, coaching decisions, team context, contract structure, and opportunity variance — is not captured by our rookie-season variables.

A third limitation is **residual left-truncation**: draft-age SD in our incident cohort (2.40) is close to the true NBA rookie-age range ($\approx$1.5–1.7) but not fully there — our filter only excludes players whose first recorded row is the dataset's first season, missing players who took a multi-year gap before re-entering. A stricter filter cross-referencing `draft_year` (already present in the raw data) could resolve this fully. Finally, players are merged by `player_name` (no unique ID exists); we found no evidence of name-collisions, but note it as a data-quality caveat.

B. Future Work

The censoring issue above points naturally toward **survival analysis** as the statistically appropriate framework for this question: methods such as the Kaplan–Meier estimator or a Cox proportional-hazards model are explicitly designed to handle right-censored time-to-event data, and would allow still-active players' careers to be included without treating their current career length as final. Our empirical hazard curve (Fig. 2) already points in this direction directly: its U-shaped pattern — elevated exit risk in the first 1–2 seasons, a mid-career trough around seasons 5–7, and rising risk again in the later seasons — suggests distinct underlying mechanisms at different career stages (unproven rookies being cut early vs. aging veterans retiring late), which a full Kaplan–Meier estimate would characterize more rigorously, with proper handling of the right-censored (still-active) players our current analysis excludes. Beyond this: a stricter incident-cohort filter using `draft_year` could resolve the residual left-truncation noted above, and the GLRT comparison could extend to a three-way decomposition isolating physical attributes separately.

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